Berkay ARDA

Full-Stack Software Developer - Computer Engineer

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EXPERIENCE

Teracity Software Technologies, Computer Engineering Intern — August 2024

- Unity · C# · Game Development · Object-Oriented Programming · UI Design
- Developed a 2D action-adventure game using Unity and C#, focusing on modular design and gameplay mechanics.
- Implemented core systems including inventory, enemy AI, wave-based spawning, score management, and UI transitions.
- Applied object-oriented programming and Singleton to create scalable and reusable class structures.
- Collaborated with other interns under supervision to review and improve and optimize project performance and usability.

EDUCATION

Izmir University of Economics, Computer Engineering (%100 English, GPA: 3.20) — 2020 - 2025

- Co-authored an academic paper accepted for presentation at ISPA 2025 – 14th International Symposium on Image and Signal Processing and Analysis, co-sponsored by IEEE Signal Processing Society and EURASIP; research based on the *ShielDir* AI-powered real-time threat detection system.
- Completed multiple team-based software and engineering projects, focusing on collaboration, version control and agile development practices.

Link: ShielDir Conference Paper

Izmir University of Economics, M.Sc. Computer Engineering (%100 English) — 2025 - Ongoing

- Conducting thesis research on reinforcement learning-based optimization and data augmentation techniques, focusing on improving AI model performance and decision-making processes.
- Developing an AI-integrated Wireless Sensor Network (WSN) framework with blockchain support for secure, energy-efficient, and tamper-resistant data transmission.

SKILLS

- Java, Python, JavaScript, C, C++, C#, SQL, HTML/CSS, PHP
- React.js, Flask, .NET, ASP.NET, ASP.NET Core
- Multithreading, RESTful APIs, API Integration
- PostgreSQL, Firebase, SQL, Database Design
- Python (for ML), Predictive Modeling
- Data Analysis, TensorFlow, PyTorch, YOLO, OpenCV, Computer Vision, Deep Learning, Model Deployment, Real-Time Inference
- Unity (2D/3D Game Design, C# Scripting, AI Mechanics, Physics System, UI/UX, Animation), TextMesh Pro, NavMesh, Particle System
- Git, GitHub, Agile Methodology, Project Management

PROJECTS

ShielDir: AI-Powered Real-Time Threat Detection System

- Python · YOLOv8–v12 · VideoMAE · OpenCV · TensorFlow · PyTorch · Firebase· Docker
- Built an AI-based surveillance system integrating YOLOv11 for weapon detection and VideoMAE for violent behavior analysis.
- Focused on low latency, scalability, and ethical AI design for public safety applications.
- Designed and containerized a real-time alert system using Docker to ensure consistent deployment, scalability, and secure data handling for mobile notifications.
- Trained custom models on a CCTV dataset using Roboflow and Google Colab, achieving high detection accuracy.

Virtual Event Management System

- C# · WinForms · ASP.NET Core · REST API · SQL
- Designed and built a full-stack web and desktop solution for managing virtual events, with real-time notifications, registration tracking, and event filtering, both for desktop and web.
- Implemented role-based access control and connected all components via RESTful APIs with a SQL Server backend.

Smart Wildfire Management

- Python · Machine Learning · Bioengineering · Random Forest Algorithm
- Collaborated in a multidisciplinary team to develop an AI-driven wildfire early-warning system.
- Utilized Random Forest models to analyze environmental and vegetation data for wildfire risk prediction.
- Experimented AI-based risk analysis with genetically engineered fire-resistant tree species.
- Designed a smart monitoring infrastructure using thermal imaging cameras and drone surveillance.
- Incorporated detection of early danger signals such as heat, smoke, and anomalous wildlife behavior.

EasyAssign – Automated Coding Assignment Evaluation Tool

- C# · WinForms · .NET · SQL
- Developed a lightweight desktop tool to help educators evaluate coding assignments faster across multiple programming languages.
- Implemented automatic ZIP file processing for student submissions, enabling bulk compilation, execution, and grading.
- Designed a configuration management system allowing users to create, import, and export language-specific build environments.
- Integrated output comparison to match student program results with expected outputs for instant grading accuracy.
- Added robust error handling to detect and report compilation or runtime errors without halting batch evaluation.

Kormos – 3D Supernatural Detective Adventure Game

- Unity · C# · 3D Game Development · AI · Animation · UI Design
- Designed and developed a 3D adventure game featuring a supernatural detective solving dark mysteries through action, puzzles, and tactical combat.
- Implemented unique player abilities such as Clone Summoning (multi-target attack system) and Time Manipulation (slow-motion combat control).
- Created enemy AI including Rats (close-range chasers), Priests (ranged attackers), and a boss enemy Rat King with summoning mechanics.
- Built an interactive UI system displaying player health, ammo, and crosshair feedback for real-time targeting.
- Integrated 3D models, animations, and effects using Unity's prefab and animation trigger systems for optimized asset management.

CERTIFICATES

- Advanced Java Program Development | Mustafa Murat Coşkun 2023
- Python Programming and Cyber Security | Atıl Samancıoğlu 2023

LANGUAGES

- **English:** Professional Working Proficiency
- **Spanish:** Elementary Proficiency
- **Turkish:** Native Language